Exam.Code:0505
Sub. Code: 3871

2122

Master of Computer Applications
First Semester
CS-2110: Data and File Structures Using C

Time allowed: 3 Hours **Max. Marks: 80**

NOTE: Attempt <u>five</u> questions in all, including Question No. 9 (Section-E) which is compulsory and selecting one question each from Section A-D.

x-x-x

SECTION-A

1)
a) Explain the concept of local/global variables, and data type, with the help of an example of each, in the context of 'C' programming language. How can the number of bytes allocated to each data type be determined for a particular 'C' compiler?
b) State one similarity and one difference between *while*, *do-while* and *for* loops in 'C' by taking examples. Explain the utility of *break* and *continue* statements in these loops. (8, 8)

2)
a) What is a function and function call in 'C'? What is the difference between call by value and call by reference in passing arguments to a function? Explain by taking suitable examples.
b) What do you mean by recursive function? Consider the following recursive 'C' function that takes two unsigned integer arguments:

```
unsigned int F(unsigned int N, unsigned int R)
{
    If (N>0) return ((N % R) + F(N / R, R));
    else return 0;
}
```

i) What is the return value of the function F when it is called as F(345,10)?
ii) What is the return value of the function F when it is called as F(513,2)? (8, 4, 4)

SECTION-B

3)
a) Discuss any four string handling built-in functions in by taking examples. Write a function in 'C' to find the length of a string without using any built-in function.
b) Compare and contrast the *struct* and the *union* as it is defined in 'C'. Explain the difference between the "→" and "." symbols as used to access members of structures. (8, 8)

4) Write a 'C' program that makes a backup file. The program should repeatedly prompt for the name of a file to backup until a name is provided that corresponds to an available file and then it should prompt for the name of the backup file and create a file copy. (16)

SECTION-C

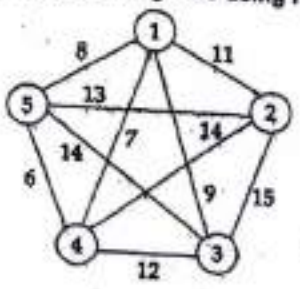
5)
a) Explain an algorithm for Quick Sort and trace it for the items: 58, 22, 15, 6, 45.
b) What is linked list? Compare linked list and linear list. Give an algorithm to insert an element at the end of a linked list. (8, 8)

6)
a) Write C-functions to implement recursive versions of pre-order, in-order and post-order traversals of a binary tree.
b) Define a stack. Explain the operations that can be performed on a stack. Write an algorithm for pushing an element into a stack. (8, 8)

P.T.O.

SECTION-D

7) Explain PRIM'S algorithm to find Minimum Cost Spanning Tree. What are its applications? Draw the Minimum Cost Spanning Tree using Prim's Algorithm for the Graph given below:



8) (16)

a) What do you mean by file organization? Differentiate between sequential, hashed and random file organization.

b) What is aim of hashing? What are different hashing techniques? What are the problems encountered in them and how to overcome them. (8, 8)

SECTION-E

9)

a) What is type casting in 'C'? What is its need? Explain with appropriate examples.

b) Given the variable declarations *int i, *ip;* and the assignment *ip=&i;* state the type of *ip* and explain the meaning of the following three expressions: *&ip, ip, *ip.*

c) A queue is an example of a _____ in _____ out data structure, whereas a stack is an example of a _____ in _____ out data structure.

d) Construct a B-Tree of order 3 for the following set of input data:

69, 19, 43, 16, 25, 40, 132, 100, 145, 7, 15, 18 (4, 4, 4, 4)

x-x-x

Exam.Code:0505
Sub. Code: 3872

2122

Master of Computer Applications

First Semester

CS-2111: Computer Organization and Architecture

Time allowed: 3 Hours Max. Marks: 80

***NOTE**: Attempt five questions in all, including Question No. IX (Unit-V) which is compulsory and selecting one question each from Unit I - IV. All questions carry equal marks.*

x-x-x

UNIT - I

I. a) Give the design of a Full adder circuit.

b) Explain and draw the characteristic table of JK flip flop.

II. a) Write and draw the truth table for different logic microoperations.

b) What is Register Transfer language? Discuss.

UNIT - II

III. a) What is an instruction cycle? Differentiate between one, two and three address instruction codes.

b) Differentiate between hardwired and microprogrammed control approach.

IV. a) Compare one-pass and two-pass assembler.

b) Give the structure of 8086/8088 CPU.

UNIT - III

V. a) What is memory hierarchy? Cost per bit of storage increases as we move up the hierarchy. Comment

b) Explain the working of associative memory.

VI. a) Give salient features of vector processing.

b) How inter-process communication takes place.

UNIT - IV

VII. a) Differentiate between SIMD and MIMD.

b) What is a superscalar design?

P.T.O.

VIII. a) Write characteristics of loosely coupled microprocessors.

b) Give description of any one algorithm for array processors.

UNIT - V

IX. Attempt the following:-

a) What are don't care conditions?

b) What is a multiplexer?

c) What are mnemonics?

d) Define term "synchronization".

e) What is hit ratio?

f) What are zero-address instructions?

g) Define term "parity".

h) What is the task of execution unit (EU)?

x-x-x

Exam.Code:0505
Sub. Code: 3873

2122

Master of Computer Applications

First Semester

CS-2112: Mathematical Structures and Numerical Techniques

Time allowed: 3 Hours | Max. Marks: 80

***NOTE**: Attempt five questions in all, including Question No.9 (Section-E) which is compulsory and selecting one question each from Section A-D. Use of calculator is allowed.*

x-x-x

SECTION A

1.) (a) Define pigeonhole principle and justify atleast two persons birthday will be in same month out of 15 persons.

(b) Arrange in ascending order for $O(n^2), O(n\,log(n)), O(1), O(n^3), O(2^n), O(log(n)), O(n)$.

(c) List two examples each for monotonic and non-monotonic functions.

(6+4+6=16)

2.) (a) Show that if any 11 numbers are chosen from the set {1,2, ...,20}, then one of them will be multiple of another.

(b) List examples for functions, one-to-one, onto and defined everywhere.

(c) Define three asymptotic notations.

(6+6+4=16)

SECTION B

3.) (a) Give truth tables for P and Q, implication, converse, inverse and contrapositive.

(b) Prove that the expressions $((A \wedge B) \leftrightarrow A)$ and $((\sim A \wedge (\sim B \wedge \sim C)) \vee (B \wedge C) \vee (A \wedge C)))$ are tautologies or not? (8+8=16)

4.) (a) Differentiate between propositional and predicate logic with examples.

(b) Prove that, $P \rightarrow (Q \vee R) \Leftrightarrow (P \rightarrow Q) \vee (P \rightarrow R)$ and $\neg(P \leftrightarrow Q) \Leftrightarrow (P \vee Q) \wedge \neg(P \wedge Q)$.

(8+8=16)

SECTION C

5.) Find roots using bisection and newton-raphson methods for $x^3 - 2x - 5 = 0$.

(8+8=16)

6.) (a) Solve using Guass Siedal method for $2x - y = 7, -x + 2y - z = 1, -y + 2z = 1$.

(b) Write formula to find next point in false-position method. How it is different from secant method?

(8+8=16)

SECTION D

7.) (a) Find $f(x)$ at 15 using newton forward and at 45 using newton backward difference methods for the data,

x	11	21	31	41	51
$f(x)$	46	66	81	93	101

(b) Apply Trapezoidal rule to evaluate $\int_0^1 \frac{2}{1+x^2}\,dx$ using 8 intervals. (8+8=16)

8.) (a) Find straight line that fit data,

x	1	2	3	4	6	8
y	2.4	3	3.6	4	5	6

(b) Apply Simpson's 1/3 and 3/8 rules to evaluate $\int_0^1 \frac{1}{1+x^2}\,dx$ using 6 intervals.

(8+8=16)

SECTION E

9.) (i) What are negations of the statements in predicate form:

(a) "All drivers are punctual" (b) $\exists x\,(x^2 > x)$

(ii) If, $G(x) = g_0 + g_1x + g_2x^2 + \cdots$ then find, $(1,1,1, \ldots)$ and $(1, -1,1, -1, \ldots)$

(iii) Why of rate of convergence important for techniques to find roots of non-linear equations?

(iv) How regression analysis different from interpolation?

$(4 \times 4 = 16)$

x-x-x

Exam.Code:0505
Sub. Code: 3874

2122

Master of Computer Applications

First Semester

CS-2113: Relational Database Management System

Time allowed: 3 Hours

Max. Marks: 80

***NOTE**: Attempt <u>five</u> questions in all, including Question No. IX (Unit-V) which is compulsory and selecting one question each from Unit I-IV*

x-x-x

UNIT - I

I. Define Normalization. Draw the difference between 4NF and 5NF with suitable examples. (16)

II. What do you mean by Data Independence? Draw the difference between Physical data independence and logical data independence. (16)

UNIT - II

III. What do you understand by Relational calculus? Draw the difference between Tuple oriented and Domain oriented relational calculus. (16)

IV. What is Concurrency control? Write down the use of Locking techniques in Concurrency control. (16)

UNIT - III

V. What do you understand by GROUP BY? Describe the use of ROLLUP and CUBE with suitable examples. (16)

VI. What is Join? Explain the different types of joins with practical examples. (16)

UNIT - IV

VII. What do you mean by Loop? Describe different types of loops in PL/SQL with suitable examples. (16)

VIII. What do you mean by Package? How can we create packages in PL/SQL? (16)

P.T.O.

UNIT - V

IX. Write short notes on following:-

a) Cursor

b) Cross Join

c) Integrity constraints

d) Log based recovery (4x4)

x-x-x

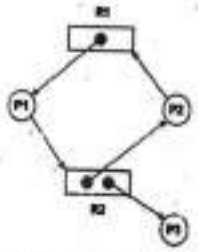
Exam.Code:0505
Sub. Code: 3875

2122
Master of Computer Applications
First Semester
CS-2114: Operating Systems

Time allowed: 3 Hours | Max. Marks: 80

***NOTE**: Attempt five questions in all, including Question No. IX (Unit-V) which is compulsory and selecting one question each from Unit I – IV.*

X-X-X

	Unit-I	
I	Consider three processes, all arriving at time zero, with total execution time of 10, 20 and 30 units respectively. Each process spends the first 20% of execution time doing I/O, the next 70% of time doing computation, and the last 10% of time doing I/O again. The operating system uses a shortest remaining compute time first scheduling algorithm and schedules a new process either when the running process gets blocked on I/O or when the running process finishes its compute burst. Assume that all I/O operations can be overlapped as much as possible. For what percentage of does the CPU remain idle?	16
II	a. What is dining philosopher's problems? How operating system does solve this problem? Discuss with the help of examples? b. Discuss he disadvantages of real time operating systems and parallel operating system?	16
	Unit-II	
III	Consider the resource allocation graph in the figure given below and Find if the system is in a deadlock state otherwise find a safe sequence? 	16
IV	a. In a virtual memory system, size of virtual address is 32-bit, size of physical address is 30-bit, page size is 4 Kbyte and size of each page table entry is 32-bit. The main memory is byte addressable. Which one of the following is the maximum number of bits that can be used for storing protection and other information in each page table entry? b. A system uses 3 page frames for storing process pages in main memory. It uses the First in First out (FIFO) page replacement policy. Assume that all the page frames are initially empty. What is the total number of page faults that will occur while processing the page reference string given below- 4, 7, 6, 1, 7, 6, 1, 2, 7, 2 Also calculate the hit ratio and miss ratio?	16
	Unit-III	
V	a. Why Linux is considered more secure operating system Justify with the help of examples? b. What is the complete process of password generation in Linux Operating System?	16
VI	a. Consider a file named EMP that contains N records (each contain 6 field separated with a | symbol) in the following form: **5678|SUMIT|DIRECTOR|SALES|19/01/43|160000** Where 5678, SUMIT, DIRECTOR, SALES, 19/01/43 and 160000 are the ID, Name, Designation, Department, Hire date and Salary of the employee respectively. Write simple Linux commands : • To count the employees under each department? • To display the employee who has the third highest salary? • To count records alphabetically (first character of name of employee)? • To display the department which has the lowest salary? • To display the eldest employee? b. Write a shell script that display the sum of 20 numbers sent as command line arguments?	16
	Unit-IV	
VII	a. What are the various files that are used by the administrator to manage groups and users? b. What is the boot sequence of a Linux operating system? What happens if the system halts abnormally? What are the commands used by the system administrator to boot a Linux system after an abnormal termination?	16

P.T.O.

VIII	Discuss the various input and output related system calls in Linux with the help of examples?	16
	Unit-V	
IX	a. Calculate the size of memory if its address consists of 22 bits and the memory is 2-byte addressable? b. Which character is used the default case in *case* construct in Linux? c. Consider a machine with 64 MB physical memory and a 32 bit virtual address space. If the page size is 4 KB, what is the number of bits in physical address? d. Which 2 commands can be used to restart Linux operating system? e. A good page replacement algorithm is one that minimizes the number of page faults? f. What are the primary options in *tar* command? g. What is the need of process synchronizations? h. What is the PID of *init* process?	16

X-X-X

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2123

Master of Computer Applications

First Semester

CS-2110: Data and File Structures Using C

Time allowed: 3 Hours | Max. Marks: 80

NOTE: Attempt five questions in all, including Question No. 9 (Section-E) which is compulsory and selecting one question each from Section A-D.

x-x-x

SECTION-A

1)

a) Explain scope and life time of variables in 'C' by taking appropriate program examples.

b) Write a function in 'C' which takes the address of a single dimensional array (containing a finite sequence of numbers) and the number of numbers stored in the array as arguments and stores the numbers in the same array in reverse order. (8, 8)

2)

a) Explain formatted input and output statements used in 'C' with syntax and examples.

b) Explain the function and need of each of the bitwise operators: &, |, ^, ~. Illustrate each operator with an example. (8, 8)

SECTION-B

3)

a) What is string in 'C'? Write a 'C' program that reads a sentence and prints the frequency of each of the vowels and total count of consonants.

b) What is pointer in 'C'? Explain how the pointer variables declared and initialized? Explain the array of pointers in 'C' with example. (8, 8)

4)

a) What is union in 'C'? How it is different from structure in 'C'? How a union is declared in 'C'?

b) What is the output of the following 'C' program? (Assuming little – endian representation of multi-byte data in which Least Significant Byte (LSB) is stored at the lowest memory address.)

```
#include<stdio.h>
#include<stdlib.h>
/* Assume short int occupies two bytes of storage */
int main ()
{
    union saving
    {
       short int one;
       char two[2];
    };
    union saving m;
    m.two [0] =5;
    m.two [1] =2;
    printf("%d, %d, %d\n", m.two[0], m.two[1], m.one);
}/* end of main */
```

(8, 8)

SECTION-C

5)

a) What is binary search? How does it work? What are some differences between sequential search and binary search? Explain with examples.

b) Briefly explain how Merge Sort works. List important advantages of Merge Sort, compared with other sorting algorithms. (8, 8)

P.T.O.

6)

a) What is a linear queue? Write an algorithm for adding an element to a linear queue. How is circular queue better than linear queue?

b) Write a recursive C (or C-like) function max-min to find the largest and the smallest numbers in a set of N different integers stored in an array A. (8, 8)

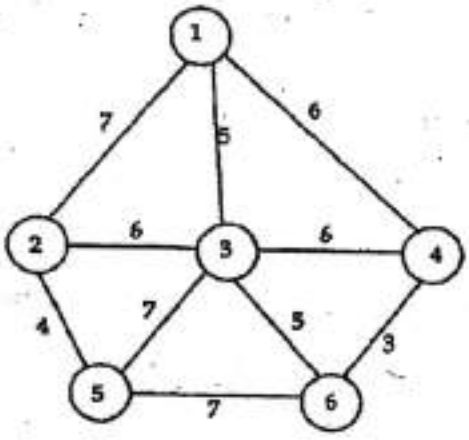
SECTION-D

7)

a) What are the differences between sequential and direct file organization? Under what conditions, if any, is it advantageous to have the file organized as a direct file rather than sequential file?

b) What are various applications of a binary search tree? Create a Binary Search Tree for the following data and perform In-order, Preorder and Post-order traversal of the tree: 40, 65, 25, 55, 10, 70, 30, 50, 15, 80, and 75. (8, 8)

8) Define spanning tree and its applications. Find the minimum spanning tree for the graph given below:



(16)

SECTION-E

9)

a) Explain the syntax and importance of *automatic* and *static* as types of storage classes in 'C' with examples.

b) In 'C' programming language, how do you read from a file and write into a file? Explain.

c) Doubly linked list takes more space than singly linked list for storing one extra address. In what condition a doubly linked list could be more beneficial than singly linked list?

d) What is aim of hashing in the context of direct file organization? What are different hashing techniques? (4, 4, 4, 4)

x-x-x

Exam.Code:0505
Sub. Code: 3872

2123

Master of Computer Applications

First Semester

CS-2111: Computer Organization and Architecture

Time allowed: 3 Hours Max. Marks: 80

***NOTE**: Attempt five questions in all, including Question No. IX (Unit-V) which is compulsory and selecting one question each from Unit I -IV.*

x-x-x

UNIT - I

I. Justify that truth tables and K-maps are the building blocks of designing functionality of a computer system. Support your answer using suitable examples of your own choice. (16)

II. Distinguish the role of combination and sequential logic design with illustrative circuit diagram. Write steps involved to design Mod-N counter and write about its real time applications. (16)

UNIT - II

III. Describe various logic and shift micro operations and their relevance in microprocessor programming. (16)

IV. Discuss various instruction codes and instruction formats available for writing programs in machine language. (16)

UNIT - III

V. Write note on:-

a) Types of interrupts

b) Types of memories (16)

VI. Draw the pin diagram of 8086/8088 CPU elaborating the usage of each pin in the microprocessor architecture. (16)

UNIT - IV

VII. Distinguish between:-

a) RISC and CISC

b) SIMD and MIMD (16)

P.T.O.

VIII. How is vector processing different from array processing? Outline applications of inter-processor communication and synchronization. (16)

UNIT - V

IX. Attempt the following:-

a) Compare working of encoder and decoder. (3)

b) Give an application of ripple counter. (3)

c) Write about and instruction cycle. (3)

d) What is the relevance of DMA? (3)

e) Define RTL. (2)

f) Why is super scalar pipeline design implemented? (2)

x-x-x

Exam.Code:0505
Sub. Code: 3874

2123

Master of Computer Applications

First Semester

CS-2113: Relational Database Management System

Time allowed: 3 Hours | Max. Marks: 80

***NOTE**: Attempt <u>five</u> questions in all, including Question No. IX (Unit-V) which is compulsory and selecting one question each from Unit I -IV.*

x-x-x

UNIT - I

I. Discuss the three tier architecture of database system in detail highlighting the role of data independence in the same. (16)

II. Justify that the concept of Normalization is the backbone of database management system. Explain any four normal forms with suitable example. (16)

UNIT - II

III. Discuss the following operations with illustrative examples:-

a) Union and Intersection

b) Projection and Selection (16)

IV. Explain the relevance of ACID properties and states of transaction. Outline various techniques followed in DBMS for concurrency control. (16)

UNIT - III

V. Describe various types of statements available in SQL. Write a note on pattern matching functions with sample SQL queries. (16)

VI. Write SQL commands with illustrative examples for:-

a) Creating table from existing table

b) Adding constraints with attributes

c) Inserting values in a table

d) Viewing data from multiple tables (16)

P.T.O.

UNIT - IV

VII. Discuss PL/SQL architecture with its major components. Elaborate conditional and iterative controls available in SQL with suitable examples. (16)

VIII. Write note on:-

a) Block structure of PL/SQL

b) User defined and predefined exceptions

c) Types of triggers

d) Cursor management (16)

UNIT - V

IX. Attempt the following:-

a) Write a note on applications of database systems. (3)

b) Draw major symbols of ER model. (3)

c) Explain log base recovery. (3)

d) How is grouping of data done from tables in SQL? (3)

e) Define role of Grant and Revoke Commands (2)

f) What is composite data type in PL/SQL? (2)

x-x-x

Exam.Code:0505
Sub. Code: 3875

2123

Master of Computer Applications

First Semester

CS-2114: Operating Systems

Time allowed: 3 Hours Max. Marks: 80

NOTE: Attempt five questions in all, including Question No. IX (Unit-V) which is compulsory and selecting one question each from Unit I -IV.

x-x-x

UNIT - I

I. a) List five services provided by an operating system. Explain how each provides convenience to the users. Explain also in which cases it would be impossible for user-level programs to provide these services.

b) What is process synchronization? Describe the following problems in the context of process synchronization:

(i) Reader-Writer problem

(ii) Dining Philosophers problem (2x8)

II. a) What do you mean be process in a computer system? What are the different process states? Explain the process life cycle in an operating system.

b) Suppose that the three processes P1, P2 and P3 arrive for execution at the times indicated as (P1, 0.0, 7), (P2, 0.5, 4), (P3, 1.2, 2) where the second component of the triple is the time arrival and the third component is the burst time. Each process will run the listed amount of time. Use non-preemptive scheduling, and base all decisions on the information that you have at the time the decision must be made.

i) What is the average turnaround time for these processes with the FCFS scheduling algorithm?

ii) What is the average turnaround time for these processes with the SJF scheduling algorithm? (2x8)

UNIT - II

IiI. What is disk scheduling? How is it different from disk management? Assume that the disk head is currently at cylinder 0. (Where the total cylinders are from 0 to 199). The disk queue requests are 75, 120,155, 80, 170, 60, and 100. Calculate the total distance (in cylinders) moved by the disk arm by the SCAN, LOOK and SSTF algorithms. (16)

IV. a) Consider a paging hardware with a TLB. Assume that the entire page table and all the pages are in the physical memory. It takes 10 milliseconds to search the TLB and 80 milliseconds to access the physical memory. If the TLB hit ratio is 0.6, compute the effective memory access time (in milliseconds).

b) Consider a demand paging system with four page frames (initially empty) and LRU page replacement policy. Compute the page fault rate (defined as the ratio of number of page faults to the number of memory accesses) for the page reference string: 7, 2, 7, 3, 2, 5, 3, 4, 6, 7, 7, 1, 5, 6, 1. (2x8)

P.T.O.

UNIT - III

V. a) How is Linux shell different from Linux kernel? Elaborate the salient features of Bourne Again Shell (BASH).

b) In an ext2 Linux file system, an inode consists of 12 Direct Disk Block Addresses (DBA), 1 single indirect DBA, 1 doubly indirect DBA and 1 triple indirect DBA. The DBA requires 32 bits and the Disk block size is 1KB. Then what is the maximum file size possible? (2x8)

VI. a) Write a Bourne shell script that prints the name and size of all files in the current directory that are larger than bytes, where x is a command line argument. For example, if the program is saved to a file called 'bigfiles', and we run it is "bigfiles 4096", then it will list all files larger than 4096 bytes.

b) Write a shell script that reads the input file, translates all the characters in the input file into the uppercase, and then stores the results in the specified output file. Synopsis: upper -i infile -o outfile. (2x8)

UNIT - IV

VII. What is Operating system administration? Explain the role and functions of a system administrator in Linux. As a Linux administrator, how do you (i) administer user and group accounts (ii) perform backup and restore operations. Explain with examples. (16)

VIII. a) Write a **'C'** program to create a child process as that it can generate Fibonacci sequence using command line arguments i.e. the number of the sequence will be provided in the command line. Make sure that parent process waits for the child to execute first.

b) For each of the following pairs of terms, define each term, making sure to clarify the key difference(s) between the two terms:

i) signal() and kill()

ii) read() and writer() (2x8)

UNIT - V

IX. Attempt the following:-

a) Explain the difference between preemptive and non-preemptive (or co-operative) CPU scheduling. Which scheme is 'the best' for a general purpose OS, would you say? Justify!

b) A byte-addressable memory system has 48-bit virtual addresses and 32-bit physical addresses, with 8KB pages. Find the number of virtual pages that each process's page table has.

c) What is the importance of regular expressions in string handling? Give examples.

d) What is Linux system run level? How do you move between the run levels (including shutdown)? How do you boot your computer (running Linux) into a single user mode? (4x4)

x-x-x

Exam.Code:0505
Sub. Code: 3873

2123

Master of Computer Applications
First Semester
CS-2126: Mathematical Structures and Linear Programming

Time allowed: 3 Hours **Max. Marks: 80**

NOTE: Attempt five questions in all, including Question No.9 (Unit-V) which is compulsory and selecting one question each from Unit I-IV.

x-x-x

UNIT I

1.) (a) Write short notes for: Pigeonhole principle and asymptotic notations. (8)
(b) (i) Let $f: R \to R$ and $g: R \to R$ be defined by $f(x) = 2x + 1$ and $g(x) = x^2 - 2$. Find composition functions $g \circ f, f \circ f, g \circ g$ and $f \circ g$.
(ii) Find the minimum number of students that can be admitted to a college so that there are at least 15 students from one of the 50 states. (4+4=8)

2.) (a) Write short notes for: invertible, one-to-one, defined everywhere and onto functions using examples. (8)
(b) (i) Prove using mathematical induction, $1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$
(ii) Prove that, $1 - x + x^2 - x^3 + x^4 + \cdots = \frac{1}{1+x}$. (4+4=8)

UNIT II

3.) (a) Prove that expressions are tautologies or not for, (i) $((\sim A \wedge (\sim B \wedge \sim C)) \vee ((B \wedge C) \vee (A \wedge C)))$
(ii) $((p \to q) \wedge (q \to r)) \to (p \to r)$. (4+4=8)
(b) Prove that, $P \to (Q \vee R) \Leftrightarrow (P \to Q) \vee (P \to R)$ and $\neg(P \leftrightarrow Q) \Leftrightarrow (P \vee Q) \wedge \neg(P \wedge Q)$. (4+4=8)

4.) (a) Write eight rules of inference in logics. (8)
(b) What is truth table for: $\neg(P \vee (Q \wedge R)) \leftrightarrow ((P \vee Q) \wedge (P \vee R))$ (8)

UNIT III

5.) (a) For an undirected graph $G = (V, E)$, prove that sum of degree of vertices is equal to two times number of edges. Also prove that graph G has an even number of vertices with odd degree.
(b) A fair coin tossed three times, find probabilities for: (i) first toss is head, (ii) second toss is head.
(c) Write short note on partial ordering with examples. (6+6+4=16)

6.) (a) Write formulas for number of vertices and edges in complete graph, cycle graph, wheel graph.
(b) The probability that A hits a target is 0.25, and the probability that B hits the target is 0.4. Both shoot at the target. Find the probability that at least one of them hits the target.
(c) Find maximal, minimal, greatest lower bound, least upper bound for the set A={2,4,6,8,12,18,24,36,72}. (6+4+6=16)

UNIT IV

7.) (a) Write short note on solving LPP using graphical method.
(b) Solve by simplex method, max $Z = 2x_1 + x_2$ s.t $x_1 + 2x_2 \le 10$, $x_1 + x_2 \le 6$, $x_1 - x_2 \le 2$, $x_1 - 2x_2 \le 1$, where $x_1, x_2 \ge 0$ (6+10=16)

8.) (a) Write short note on applications of Linear programming.
(b) Solve by simplex method, max $Z = 3x_1 + 2x_2 + 5x_3$ s.t. $x_1 + x_2 + x_3 \le 9$, $2x_1 + 3x_2 + 5x_3 \le 30$, $2x_1 - x_2 - x_3 \le 8$, where $x_1, x_2, x_3 \ge 0$ (6+10=16)

UNIT V

9.) (i) What are negations of the statements in predicate form:
(a) "All drivers are punctual" (b) $\exists x\,(x^2 > x)$
(ii) Give two examples each for monotonically increasing and decreasing functions.
(iii) Write formulas for expected values and variance.
(iv) List assumptions in linear programming models.
$(4 \times 4 = 16)$

x-x-x